

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-acb4e457b5bacb733.jsonl`
- SHA-256: `b8a9cf0925584518b7591a9502f89d513a2fa57432bd45ef91507459a0641d9e`
- Source modified: `2026-07-03T16:42:16+00:00`
- Imported at: `2026-07-05T16:48:26+00:00`
- project: `subagents`
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-03T16:40:18.318Z)

You are a code-review finder agent. Review target: PR #451 of the badminton-highlight-indexer repo, checked out at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer on branch codex/alpha-readiness-plan (HEAD cd07339). Get the diff with: git diff origin/master..HEAD. Docs-only PR: new docs/ALPHA_LAUNCH_READINESS.md plus index additions.

ANGLE ? efficiency/process-waste. For a docs/process PR this means: does the §7 tracker sequence work inefficiently ? serialized deliverables that are actually independent (a Depends-on column entry that isn't a real dependency, forcing needless serialization), duplicated verification work across rows, or a §8 PR sequence that re-runs expensive steps (GPU runs, feature regeneration) more times than needed? Also check the §11 verification commands: are any of them wrong/won't-run as written against the current repo (flag only if you verify the entryptpoint/flag doesn't exist ? check backend/eval/*.py, training/gen0/harness.py, scripts/nightly_regression.py argument parsers for the flags used: --features-dir, --manifest, --floor, --version, --out, --no-report, --granularity group)?

Return up to 6 candidate findings as a JSON array of {file, line, summary, failure_scenario}. Return [] if none ? this angle may legitimately find nothing. Your final message must be only the JSON array.

2. assistant (2026-07-03T16:40:20.178Z)

I'll analyze the PR #451 diff to review the docs/process for efficiency issues in the tracker sequence and verification commands.

3. user (2026-07-03T16:40:21.085Z)

```
diff --git a/docs/ALPHA_LAUNCH_READINESS.md b/docs/ALPHA_LAUNCH_READINESS.md
new file mode 100644
index 0000000..4b3d49d
--- /dev/null
+++ b/docs/ALPHA_LAUNCH_READINESS.md
@@ -0,0 +1,171 @@
+# Alpha Launch Readiness Plan
+
+> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:** 2026-07-03; **Last updated:** 2026-07-03
+> **Lifecycle:** `DRAFT -> IN PROGRESS -> DONE -> archived`
+> **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed in `docs/README.md` and `docs/DOC_STATUS.md`.
+> **Relation to existing docs:** consolidates the launch-relevant pending items from `NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `kheldutra` Studio docs.
+> **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not legal, financial, or launch approval.
```

```

+
+---
+
+## 0. TL;DR
+
+The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but
the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15
badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven
commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]`
the n=15 quality baseline is stale and must be intentionally refreshed before it can gate
anything.
+
+The first controlled alpha should be a hosted, authenticated product loop: upload a video,
process it, select rallies, compile, and download a reel. The owned-model/commercial rally-
detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB
checkpoint provenance is resolved.
+
+## 1. Problem & goal
+
+The previous launch blockers were spread across roadmap docs and mixed together three different
questions:
+
+- **Can an invited tester use the product without SSH?**
+- **Can the 15-video golden corpus now support honest quality gates?**
+- **Can we claim or ship an owned/commercial rally detector?**
+
+This doc separates those into executable workstreams. The goal is to reach an invite-only alpha
where the product is usable end-to-end, the quality claims are honest and reproducible, and the
remaining model/legal gates are explicit rather than accidentally implied by launch activity.
+
+## 2. Decisions locked
+
+| # | Decision | Source / date | Implication |
+|---|---|---|---|
+| D1 | **Product alpha and owned-model launch are separate decisions.** | `NEXT_STEPS.md` NO-GO
banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without
claiming the owned model is commercially cleared. |
+| D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
+| D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
+| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
windowing safety; keep it as an opt-in/analysis lever. |
+| D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.**
| sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use
manual/offline corpus promotion; automated flywheel follows after serving and gates are stable.
|
+
+## 3. Verified snapshot, 2026-07-03
+
+All five Git checkouts were fast-forwarded before analysis:
+
+| Repo | Branch | Synced HEAD |
+|---|---|---|
+| `badminton-highlight-indexer` | `master` | `e4914d1` |
+| `khehsutra` | `master` | `b0f70a5` |
+| `rally-annotator` | `main` | `ef9e088` |
+| `sports-data-collector` | `master` | `bfc014f` |
+| `sports-obsreport` | `main` | `a0c7252` |
+
+Corpus/eval facts:
+

```

```

+- `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says `output/human_lovo_manifest.json` is
the ingested golden corpus at `n = 15`.
+- `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found
all 15 trajectories and labels in this workspace.
+- `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
+- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only 6
videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
+- `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
+- `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
+- `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15> --features-dir
output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off tolF1 `0.1382`,
but marked the frozen baseline stale due to config/corpus changes.
+- `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity group`
currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
+
+### 4. Launch strategy
+
+### 4.1 Alpha lane: make the product usable
+
+The alpha lane is successful when an invited tester can open the app, authenticate, upload a
video, watch the job progress, pick rallies, compile a reel, and download it. This lane should
avoid model-quality promises beyond the measured state and should default to the most reliable
serving path.
+
+### 4.2 Quality lane: make the 15-video corpus gate-ready
+
+The quality lane is successful when the 15-video corpus is portable, reproducible, feature-
complete, and backed by refreshed baselines. Only then should R2 tolF1, Gen-0/TemporalMaxer, or
other model/promotion decisions be ratified.
+
+### 4.3 Flywheel lane: make tester feedback useful
+
+The flywheel lane is successful when user corrections can flow into the collector/vault path
with consent/provenance and a visible re-eval delta. This is valuable for alpha learning, but it
should not block the first controlled alpha if manual promotion can bridge the gap.
+
+### 5. Risks and guardrails
+
+| Risk | Why it matters | Guardrail |
+|---|---|---|
+| Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
+| Stale local manifests create false confidence | Manifest-driven tools skipped or failed until
paths were rebased | Make the corpus manifest portable and add a path/fixture smoke check |
+| 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still n=6
today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
+| Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly reported
stale baselines | Refresh baselines only after reviewer sign-off and record the command/output |
+| P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but can
consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha lane is
already stable |
+| WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained model
regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product launch
tasks |
+
+### 6. Honest limits
+
+Completing this doc's alpha lane does not prove the owned model is ready, that WASB weights

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are commercially clean, that the product can serve unbounded public traffic, that multi-sport is ready, or that every tester correction automatically improves the model.

+
+Completing the quality lane does ****not**** by itself deploy a service. It only makes the measurement surface honest enough for a launch decision.

+
+## 7. Deliverables & progress tracker

+
+Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a cross-repo batch.

+
+| ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
+|----|-----|-----|-----|-----|-----|----|
+| A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer` | - | No | IN PROGRESS | #451 |
+| A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelesutra` + `badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
+| A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No | TODO | - |
+| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` + vault | A2 | Owner upload/storage scope | TODO | - |
+| A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2 | Reviewer sign-off | TODO | - |
+| A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic\_lovo\_n6` floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
+| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
+| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data availability | TODO | - |
+| A8 | Re-run `fusion\_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips | `badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
+| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner approval for M0.4 scope | TODO | - |
+| A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-indexer` | A4, A5, A8 | Owner decision | GATED | - |
+| A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance | GATED | - |
+| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train|reeval` job, and delta-F1 surface | `khelesutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
+| A13 | Alpha launch review: runbook, privacy/consent copy, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch decision | TODO | - |

+
+## 8. Concrete next PR sequence

+
+1. ****Docs PR:**** land this tracker.
+2. ****Corpus portability PR:**** rebase/portable manifest tooling plus missing-file smoke test.
+3. ****Hosted alpha PR/batch:**** health/auth/upload/process/job/compile/download verification and runbook updates.
+4. ****Baseline PR:**** intentional n=15 nightly baseline refresh and heuristic floor refresh.
+5. ****Ablation/fusion PRs:**** fix ablation import, complete 15-video features, rerun promotion/rejection decisions.
+6. ****Launch decision PR/docs update:**** record ship-gate target, C3 status, alpha copy limits, and launch checklist.

```

+## 9. Open questions
+
+1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should
P10/P11 be reopened before any tester wave?
+2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4
VM, or another approved GPU host?
+3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains unresolved?
+4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
+5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip,
or should the eval loaders support mixed schemas with explicit capability tags?
+
+## 10. Definition of done
+
+- [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.
+- [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
+- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale
local absolute paths.
+- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
+- [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
+- [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected on the
15-video corpus.
+- [ ] Ship-gate target and allowed alpha claims are recorded.
+- [ ] WASB C3 status is recorded and reflected in alpha messaging.
+- [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-
plus.
+- [ ] Owner records GO/NO-GO for the first alpha wave.
+
+## 11. References
+
+Internal docs:
+
+- `docs/NEXT_STEPS.md`
+- `docs/ROADMAP.md`
+- `docs/data_pipeline/GOLDEN_VIDEOS.md`
+- `docs/R2_EVAL_METRIC_DESIGN.md`
+- `docs/RALLY_DETECTION_FIXES_PLAN.md`
+- `docs/COMPUTE_DECOUPLED_SERVING/README.md`
+- sibling `khesutra/NEXT_STEPS.md`
+- sibling `khesutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
+- sibling `khesutra/docs/PER_RALLY_SELECTION.md`
+- sibling `sports-data-collector/data/golden_source_videos.json`
+
+Local verification commands used for the 2026-07-03 snapshot:
+
+```powershell
+python -m backend.eval.fusion_compare --features-dir output
+python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
+python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
+python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir output
--no-report
+python -m backend.eval.ablation --features-dir output --granularity group
+```
+
+---
+
+#### Changelog
+
+- 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`, golden-corpus
docs, local eval artifacts, and the current 15-video corpus state.
diff --git a/docs/DOC_STATUS.md b/docs/DOC_STATUS.md
index 34411fd..2541672 100644
--- a/docs/DOC_STATUS.md

```

```

+++ b/docs/DOC_STATUS.md
@@ -30,6 +30,7 @@ Before a foundational/tracked doc moves to `docs/archives/`:
  ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's $7 is the completeness
  source-of-truth)
  | Doc | Status | Archive-ready? | Next action |
  |---|---|---|---|
+| `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created
2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-
model/commercial rally-detection gates | No | Execute $7 in order: corpus portability + n=15
baselines, hosted alpha serving, ablation/fusion reruns, then launch G0/N0-G0 |
| `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED**
(#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights
pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual:
counsel/prices/OAuth/real-run within the D17 cap) |
| `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** ($7) | No | Drives the
R-series; complete when $7.1 single quality number is attributed |
| `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation
pending) | Promote-or-reject toF1 as the gate (ablation) ? then archive with the R-cluster |
diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
index f186ff9..a9f35a8 100644
--- a/docs/NEXT_STEPS.md
+++ b/docs/NEXT_STEPS.md
@@ -9,7 +9,7 @@

```

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## ? Prioritized pending ? all tracks
**P0 ? highest leverage / unblocks the most**
-0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA
ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
`/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
[[ui-driven-cloud-serving-gap]].
+0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA
ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
`/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
[[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:**
`docs/ALPHA_LAUNCH_READINESS.md` $7.
0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-
included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install`
? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state
/ ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b,
P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** ***(1)** P3 **GPU validation +
screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB
0-rally tuned-config blocker), ***(2)** P2b-install (uv/script + truly-local default config),
***(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud
(Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-
session-packaging-app]]`.
1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified).
M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-
OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All $8 gating Qs
locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA
(M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF
team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap**
(D17).
2. **[owner, ongoing] Grow the golden corpus ? multi-court badminton + ?3 matches/sport (TT

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first). The single highest-leverage move; the quality track *and* full-corpus training feed on
it.
diff --git a/docs/README.md b/docs/README.md
index 0df1a69..58be5ff 100644
--- a/docs/README.md
+++ b/docs/README.md
@@ -15,6 +15,7 @@
- [DOC_STATUS.md](DOC_STATUS.md) ? **? living Documentation Status & Archival Register**: every
foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence
checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here;
update rows as docs complete.

## Product / platform plans
+- [ALPHA_LAUNCH_READINESS.md](ALPHA_LAUNCH_READINESS.md) ? **? Active tracked project (`DRAFT`,
created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15
videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the
owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). $7 is the
source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh,
ablation/fusion reruns, hosted serving, alpha copy/launch review.
- [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked
project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to
DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never
branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend`
(what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU,
in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run
GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI).
Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY
proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a
[runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage
ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). $7 M0?M10 tracker is the source
of truth.
- [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED +
ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing
bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB
runner**, **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ?
DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving
endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving"
subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under
`docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
- [PACKAGING_PLAN.md](PACKAGING_PLAN.md) ? **? Active tracked project (`IN PROGRESS`, 11 PRs
merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ?
`pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA**
single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision
D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md`
narrative.** $7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **?
Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4
(ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-
app]]`.

```

4. assistant (2026-07-03T16:40:23.286Z)

Now let me check the verification commands referenced in the doc against the actual repository code to identify any that won't run as written:

5. user (2026-07-03T16:40:25.117Z)

```

./backend/eval/ablation.py
./backend/eval/ablation_pdf.py
./backend/eval/annotations.py
./backend/eval/audio/e1_probe.py
./backend/eval/audio/__init__.py
./backend/eval/batch.py
./backend/eval/calibrate_local.py
./backend/eval/calibrate_wasb.py
./backend/eval/calibration.py

```

```

./backend/eval/classifier.py
./backend/eval/distill_local.py
./backend/eval/eval_partial_labels.py
./backend/eval/eval_stats.py
./backend/eval/experiment.py
./backend/eval/export_wasb_segments.py
./backend/eval/fusion_audio.py
./backend/eval/fusion_compare.py
./backend/eval/fusion_golden.py
./backend/eval/gemini_refine.py
./backend/eval/golden_fixtures.py

```

6. user (2026-07-03T16:40:27.873Z)

```

1      """Pure-CPU LOVO comparison: shuttle-only vs shuttle+person learned fusion on the golden
set.
2
3      The fusion-P3 measurement. Loads the per-video feature JSONs written by
`fusion_golden.py`
4      and runs the SHIPPED experiment harness (`experiment.compare`, honest leave-one-video-
out) ?
5      no new scoring/labeling math. The two variants differ ONLY by `feature_names` (the
harness's
6      designed seam): the person features are extra columns the learned LogisticRegression
weights,
7      so the held-out ?F1 is an honest answer to "does the person cue help?" ? not a hunch.
8
9      Run (no GPU):
10     python -m backend.eval.fusion_compare --features-dir output --record
11     """
12
13     from __future__ import annotations
14
15     import argparse
16     import logging
17     from typing import Any, Dict, List
18
19     from backend.eval import experiment
20     from backend.eval.eval_stats import paired_delta_ci
21     from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES
22
23     logger = logging.getLogger(__name__)
24
25
26     def delta_stats(res: Dict[str, Any], metric: str = "f1") -> Dict[str, Any]:
27         """Paired (by video) B?A summary for one metric: mean delta + bootstrap CI + sign
test
28         (eval_stats, C1 ? "never a bare mean"). Aligns the two variants' per-video held-out
scores
29         by video name, so the headline ?F1 comes with how-wide + did-it-win-on-enough-
videos."""
30         by_a = {p["video"]: p[metric] for p in res["variant_a"]["per_video"]}
31         by_b = {p["video"]: p[metric] for p in res["variant_b"]["per_video"]}
32         vids = [p["video"] for p in res["variant_a"]["per_video"]]
33         return paired_delta_ci([by_a[v] for v in vids], [by_b[v] for v in vids])
34
35
36     def load_videos(features_dir: str) -> List[experiment.VideoCandidates]:
37         """Load every FULL-FUSION-SCHEMA ``*_golden_features.json`` into a VideoCandidates
38         (intervals + the 10 feature columns + golden gts). The harness derives labels from
gts
39         itself.
40
41         Leaner velocity-windowing-only files (no ``shuttle``/``person`` cue dicts ? the
schema the

```



```

42     corpus-growth path persists) are SKIPPED, not crashed on, so a mixed corpus loads
the
43     fusion subset cleanly (see ``_is_fusion_schema``)."""
44     return experiment.load_golden_videos(
45         features_dir,
46         shuttle_columns=SHUTTLE_FEATURE_NAMES,
47         person_columns=PERSON_FEATURE_NAMES,
48         audio_policy="omit",
49         skip_logger=logger,
50         skip_message="fusion_compare.load_videos: skipping %s ? windowing-only schema "

```

7. user (2026-07-03T16:40:32.749Z)

```

usage: fusion_compare.py [-h] [--features-dir FEATURES_DIR] [--iou IOU]
                        [--record]

```

LOVO fusion lift: shuttle-only vs shuttle+person (golden set).

options:

```

-h, --help            show this help message and exit
--features-dir FEATURES_DIR
--iou IOU
--record              append the result to the experiment leaderboard

```

8. user (2026-07-03T16:40:35.007Z)

```

1     """The eval?serve worked example: score a cue at BOTH windowings and run the promotion
guard.
2
3     This is the reproducible proof of the foundational fix. For the Gate-A held-shuttle
guard it
4     builds candidate windows directly from each golden trajectory CSV under BOTH named
windowings
5     (:data:`backend.eval.windowing.PROPOSAL` and :data:`~backend.eval.windowing.SERVED`),
scores
6     CONTROL vs Gate-A (`net_crossings >= 2`) at each, and feeds the per-video held-out ?F1
into
7     :func:`backend.eval.promotion.promote_if_served_safe`.
8
9     It demonstrates the bug *and* the fix in one run:
10    - PROPOSAL windowing: Gate-A wins (historically **?F1 ? +0.074**, CI clears 0, 6/0
sign test).
11    - SERVED windowing: Gate-A does **not** win (historically **?F1 ? ?0.008**, real-
rally loss).
12    - The promotion guard therefore **vetoes** a promotion that would otherwise pass on
the
13    proposal numbers ? a cue cannot be promoted on a windowing production doesn't serve.
14
15    Unlike the persisted ``*_golden_features.json`` (which are PROPOSAL-windowed only), this
16    re-windows from the raw trajectory CSVs so the SERVED candidate set is genuinely
different.
17    Pure CPU, GPU-free, no Gemini, no DB. Reuses only shipped pieces (windowing, rally_gate,
18    metrics, promotion). Owner runs it against the real golden CSVs; tests exercise it on
synthetic
19    trajectories.
20
21    CLI::
22
23    python -m backend.eval.serve_contrast --manifest output/human_lovo_manifest.json
24    """
25
26    from __future__ import annotations
27
28    import argparse
29    import json

```

```

30     from typing import Any, Dict, List, Sequence, Tuple
31
32     from backend.eval import windowing as W
33     from backend.eval.calibrate_wasb import load_golden_gts
34     from backend.eval.gemini_refine import merge_overlaps
35     from backend.eval.metrics import score
36     from backend.eval.promotion import promote_if_served_safe
37     from backend.pipeline.detectors import rally_gate
38     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
39
40     Interval = Tuple[float, float]
41     IOU = 0.5
42     MIN_CROSSINGS = 2
43
44
45     def gate_a_scores_at(
46         points: List[Any],
47         gts: List[Interval],
48         fps: float,
49         frame_width: float,
50         preset: W.WindowingPreset,
51         iou: float = IOU,
52         min_crossings: int = MIN_CROSSINGS,
53         merge_gap: float = 1.0,
54     ) -> Tuple[float, float]:
55         """One video's (CONTROL F1, Gate-A F1) at ``preset``'s windowing.
56
57         Builds candidates under ``preset``, then scores two static variants vs ``gts``:
58         - CONTROL ? every candidate window, overlap-merged (the un-gated baseline).
59         - Gate-A ? keep only windows whose shuttle net-crossing count >=
60         ``min_crossings``.
61
62 """

```

9. user (2026-07-03T16:40:40.033Z)

```

usage: serve_contrast.py [-h] [--manifest MANIFEST] [--iou IOU]
                        [--min-crossings MIN_CROSSINGS] [--min-n MIN_N]
                        [--served-regress-eps SERVED_REGRESS_EPS]
                        [--json JSON]

```

Gate-A eval?serve worked example: score CONTROL vs Gate-A at the PROPOSAL and SERVED windowings, then run the served-guarded promotion gate.

options:

```

-h, --help            show this help message and exit
--manifest MANIFEST   JSON: [{name, trajectory, fps, frame_width, labels},
                        ...]
--iou IOU
--min-crossings MIN_CROSSINGS
--min-n MIN_N         promotion min-N floor
--served-regress-eps SERVED_REGRESS_EPS
                        served regression tolerance (0.0 = no served-mean drop
                        allowed)
--json JSON           also write the structured result JSON here

```

10. user (2026-07-03T16:40:42.061Z)

```

1     """Gen-0 TAL harness (M0.4): ONE command -> train -> honest LOVO eval -> ship-gate ->
artifact.
2
3     python -m training.gen0.harness --manifest output/human_lovo_manifest.json \
4         --floor eval_baselines/heuristic_lovo_n6.json --out artifacts/rally_tal_gen0
--version 0.1.0
5
6     Productionizes the rally_seq prototype per ADR-008:

```

```

7   - TRAINS on the corpus manifest (per-video trajectory + golden labels), evaluates with
8   honest leave-one-video-out (split by match, threshold tuned on train only), then fits
9   the FINAL model on all videos.
10  - Emits a versioned ARTIFACT BUNDLE ? the only thing serving may ever consume:
11      <out>/<version>/weights.json    (gitignored; classifier + frozen normalization)
12      <out>/<version>/manifest.json    (the reviewable contract: feature-schema version,
13                                          calibration, training lineage + GT hashes,
14                                          eval results, attestations, gate verdict)
15  - Runs the SHIP-GATE as pre-flight: floor comparison + a PAIRED per-video sign test
16    (at n=6 a mean gap alone is noise-ambiguous), plus the hard attestation blocks
17    (WASB C3 unresolved => commercial ship refused; explicit F1@tIoU target still
18    undefined => no self-declared victory). The product side owns final admission
19    (backend/eval/regression.py); this harness never grades itself into serving.
20
21  Training may import backend.* (one-way wall); backend must never import training.*.
22  """
23
24  from __future__ import annotations
25
26  import argparse
27  import hashlib
28  import json
29  import os
30  import re
31  import subprocess
32  import time
33  from math import comb
34  from typing import Dict, List, Optional
35
36  import numpy as np
37
38  from backend.eval.classifier import LogisticRegression
39  from backend.eval.gt_loader import load_gt_intervals
40  from backend.eval.rally_seq_proto import (
41      _seg_f1,
42      labels_for,
43  )
44  from backend.eval.rally_seq_proto import (
45      run as lovo_run,
46  )
47  from backend.eval.regression import gt_hash
48  from backend.features.rally_seq import (
49      FEAT_NAMES,
50      FEATURE_SCHEMA,
51      build_frame_arrays,
52      features,
53  )
54  from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
55
56  # Post-processing constants ? part of the model's identity, recorded in the manifest.
57  CALIBRATION_DEFAULTS = {"smooth": 7, "merge_gap": 1.0, "min_dur": 1.0}
58  THRESHOLD_GRID = [round(t, 2) for t in np.arange(0.2, 0.81, 0.05)]
59
60
61  # Repo root = two levels up from training/gen0/harness.py. A `GIT_SHA` file baked here
62  at
63  # tarball-build time is the off-box provenance fallback (see _git_sha +
64  deploy/gcp/README.md).
65  _REPO_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), "..", ".."))
66  _SHA_RE = re.compile(r"^[0-9a-f]{7,40}$")
67
68  #: Env var carrying the source commit SHA out-of-band when the run has no `.git` (a `git
69  archive`
70  #: tarball ? the GCP-L4 path). Set it on the box, or bake a `GIT_SHA` file into the
71  archive root.

```

```

68     GIT_SHA_ENV = "RALLY_GIT_SHA"
69
70
71     def _valid_sha(s: Optional[str]) -> Optional[str]:
72         """Return the normalized commit SHA if `s` looks like one (7-40 hex chars), else
73         None.
74         Guards every provenance source so a stray value (`unknown\n`, a build label) can
75         never be
76         stamped as if it were a real SHA."""
77         s = (s or "").strip().lower()
78         return s if _SHA_RE.match(s) else None
79
80     def _git_sha() -> str:

```

11. user (2026-07-03T16:40:48.976Z)

```

usage: harness.py [-h] --manifest MANIFEST [--floor FLOOR] [--out OUT]
                --version VERSION

```

Gen-0 TAL harness (M0.4): ONE command -> train -> honest LOVO eval -> ship-gate -> artifact. python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor eval_baselines/heuristic_lovo_n6.json --out artifacts/rally_tal_gen0 --version 0.1.0 Productionizes the rally_seq prototype per ADR-008: - TRAINS on the corpus manifest (per-video trajectory + golden labels), evaluates with honest leave-one-video-out (split by match, threshold tuned on train only), then fits the FINAL model on all videos. - Emits a versioned ARTIFACT BUNDLE ? the only thing serving may ever consume: <out>/<version>/weights.json (gitignored; classifier + frozen normalization) <out>/<version>/manifest.json (the reviewable contract: feature-schema version, calibration, training lineage + GT hashes, eval results, attestations, gate verdict) - Runs the SHIP-GATE as pre-flight: floor comparison + a PAIRED per-video sign test (at n=6 a mean gap alone is noise-ambiguous), plus the hard attestation blocks (WASB C3 unresolved => commercial ship refused; explicit F1@tIoU target still undefined => no self-declared victory). The product side owns final admission (backend/eval/regression.py); this harness never grades itself into serving. Training may import backend.* (one-way wall); backend must never import training.*.

options:

```

-h, --help            show this help message and exit
--manifest MANIFEST    corpus manifest JSON:
                        [{name,trajectory,fps,frame_width,labels}, ...]
--floor FLOOR          floor baseline JSON (e.g.
                        eval_baselines/heuristic_lovo_n6.json)
--out OUT              semver for the artifact bundle
--version VERSION

```

12. user (2026-07-03T16:40:51.106Z)

```

1     """Nightly rally-quality regression tripwire.
2
3     Re-scores the **golden corpus** under HEAD via :mod:`backend.eval.rally_seg_eval` (CPU,
4     seconds ? the WASB GPU trajectories are CACHED, only the windowing/scorer changes) and
5     diffs
6     the headline metrics against a FROZEN baseline
7     (`eval_baselines/nightly_baseline.json`).
8     Writes a dated report and exits non-zero on any regression beyond tolerance OR any per-
9     video
10    over-segmentation-guard regression.
11
12    This is the owner's dual-eval-set discipline made automatic (`memory/eval-dual-set-
13    regression`)
14    + `docs/R2_EVAL_METRIC_DESIGN.md` $4.5): the growing GOLDEN-finalized set is re-run

```

```

every
11     night so a *late* change cannot silently lose an *earlier* win. It is **default-
ADDITIVE** ? it
12     does NOT change the promotion gate. It is a *tripwire*, not a re-training run: no GPU,
no Gemini,
13     no WASB re-extraction (those are cached and unchanged night to night).
14
15     Three tracked configs (all re-scored every night; a clean A/B/C on the same SERVED base
knobs):
16
17     * **DEFAULT** ? the SERVED production windowing, i.e. what users get today. Catches a
regression
18     to shipped behaviour. **NOTE (post-#204, 2026-06-18):** the R1 milestone is now
flipped ON, so
19     SERVED *is* cap=6 + R4 ? "default" now equals "milestone".
20     * **MILESTONE** ? the explicit R3+R4 config (cap=6 s + rally-END inactivity trim;
21     ``docs/QUALITY_ITERATIONS.md`` R3/R4). Now == DEFAULT (production shipped it); kept
explicit so
22     the nightly still names the milestone knobs and stays meaningful if production later
diverges.
23     * **BASELINE_OFF** ? the pre-R1-flip all-OFF windowing (every quality knob forced OFF).
The
24     permanent historical floor, so each report keeps showing the delta the milestone buys
(a launched
25     win must keep standing its ground, not be taken at launch-time value).
26
27     Ground truth + trajectories are LOCAL (the manifest points at owned, minors-free
trajectory CSVs
28     outside the repo + ``output/<name>_golden_features.json``); the corpus is NOT committed.
So the
29     baseline is tied to the owner's local golden corpus and is regenerated with ``--update-
baseline``
30     whenever the corpus grows or an intended improvement lands. (A cloud agent does not
suffice ? it
31     has the repo but not the cached trajectories; hence the local Windows Scheduled Task
wiring in
32     ``scripts/nightly_regression.ps1``.)
33
34     Usage
35     -----
36     Snapshot the baseline from current HEAD (after an intended improvement or a corpus
change):
37
38         python scripts/nightly_regression.py --update-baseline
39
40     Nightly check (exit 1 on regression; writes ``output/nightly_regression/<date>.md`` +
``.json``):
41
42         python scripts/nightly_regression.py
43
44     Exit codes
45     -----
46     * ``0`` ? all tracked configs clean (no regression), corpus matches the baseline.
47     * ``1`` ? at least one regression (an aggregate metric dropped beyond tolerance, or a
per-video
48     over-seg guard increased). Takes precedence over a stale baseline.
49     * ``2`` ? operational error (manifest/corpus missing, nothing scored).
50     * ``3`` ? no baseline to compare against (snapshot one with ``--update-baseline``).
51     * ``4`` ? baseline STALE (the corpus set or a config definition changed): a refresh is
needed,
52     but no regression was found on the comparable intersection.
53     ""
54
55     from __future__ import annotations
56

```

```

57     import argparse
58     import datetime as _dt
59     import json
60     import os
61     import sys
62     from dataclasses import dataclass, field, replace
63     from typing import Any, Dict, List, Optional, Tuple
64
65     # ----- #
66     # Paths (resolved relative to the repo root, so the script is cwd-robust).
67     # ----- #
68     REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
69     if REPO_ROOT not in sys.path:
70         sys.path.insert(
71             0, REPO_ROOT
72         ) # importable as `python scripts/nightly_regression.py` from anywhere
73     DEFAULT_BASELINE = os.path.join(REPO_ROOT, "eval_baselines", "nightly_baseline.json")
74     DEFAULT_REPORT_DIR = os.path.join(REPO_ROOT, "output", "nightly_regression")
75     DEFAULT_MANIFEST = os.path.join(REPO_ROOT, "output", "human_lovo_manifest.json")
76     DEFAULT_FEATURES_DIR = os.path.join(REPO_ROOT, "output")
77     DEFAULT_IOU = 0.5
78
79     # ----- #
80     # What we gate on (a tripwire ? focused, not every metric).

```

13. user (2026-07-03T16:40:54.435Z)

```

usage: nightly_regression.py [-h] [--manifest MANIFEST]
                             [--features-dir FEATURES_DIR]
                             [--baseline BASELINE] [--report-dir REPORT_DIR]
                             [--iou IOU] [--metric-tol METRIC_TOL]
                             [--rate-tol RATE_TOL]
                             [--over-seg-tol OVER_SEG_TOL] [--update-baseline]
                             [--date DATE] [--no-report]

```

Nightly rally-quality regression tripwire (offline, CPU).

options:

```

-h, --help            show this help message and exit
--manifest MANIFEST    golden manifest JSON
--features-dir FEATURES_DIR
                        dir holding <name>_golden_features.json
--baseline BASELINE    frozen baseline JSON path
--report-dir REPORT_DIR
                        where dated reports are written
--iou IOU
--metric-tol METRIC_TOL
                        F1-like aggregate drop that counts as a regression
--rate-tol RATE_TOL    merge_rate rise that counts as a regression
--over-seg-tol OVER_SEG_TOL
                        per-video split/merge-count increase allowed before
                        it's a regression
--update-baseline      snapshot the current numbers as the new baseline
                        instead of checking
--date DATE            override the report date (YYYY-MM-DD; for tests)
--no-report            skip writing the dated report file

```

14. user (2026-07-03T16:40:59.619Z)

Traceback (most recent call last):

```

File "<frozen runpy>", line 198, in _run_module_as_main
File "<frozen runpy>", line 88, in _run_code
File "C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-
indexer\backend\eval\ablation.py", line 530, in <module>
    from backend.eval.ablation_pdf import export_pdf as export_pdf # noqa: E402

```



```

121    ## 9. Open questions
122
123    1. Owner: Is the first alpha allowed to use manual/offline corpus promotion, or
124    should P10/P11 be reopened before any tester wave?
125    2. Owner/platform: Which hosted compute target is the next stand-up: Cloud Run GPU,
126    GCE L4 VM, or another approved GPU host?
127    3. Owner/legal: What is the acceptable alpha wording while WASB C3 remains
128    unresolved?
129    4. Engineering: Should the portable corpus manifest live as a generated artifact, a
130    checked-in redacted manifest, or both?
131    5. Engineering: Should n=15 feature regeneration target full fusion schema for every
132    clip, or should the eval loaders support mixed schemas with explicit capability tags?
133
134    ## 10. Definition of done
135
136    - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
137    SSH.
138    - [ ] Alpha service is gated behind the chosen auth posture and has a documented
139    disable/rollback path.
140    - [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely
141    on stale local absolute paths.
142    - [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
143    - [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
144    - [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
145    on the 15-video corpus.
146    - [ ] Ship-gate target and allowed alpha claims are recorded.
147    - [ ] WASB C3 status is recorded and reflected in alpha messaging.
148    - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
149    alpha-plus.
150    - [ ] Owner records GO/NO-GO for the first alpha wave.
151
152    ## 11. References
153
154    Internal docs:
155
156    - `docs/NEXT_STEPS.md`
157    - `docs/ROADMAP.md`
158    - `docs/data_pipeline/GOLDEN_VIDEOS.md`
159    - `docs/R2_EVAL_METRIC_DESIGN.md`
160    - `docs/RALLY_DETECTION_FIXES_PLAN.md`
161    - `docs/COMPUTE_DECOUPLED_SERVING/README.md`
162    - sibling `khehsutra/NEXT_STEPS.md`
163    - sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
164    - sibling `khehsutra/docs/PER_RALLY_SELECTION.md`
165    - sibling `sports-data-collector/data/golden_source_videos.json`
166
167    Local verification commands used for the 2026-07-03 snapshot:
168
169    ```powershell
170    python -m backend.eval.fusion_compare --features-dir output
171    python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
172    python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
173    eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
174    python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir
175    output --no-report
176    python -m backend.eval.ablation --features-dir output --granularity group
177    ```
178
179    ---
180
181    ### Changelog
182
183    - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
184    golden-corpus docs, local eval artifacts, and the current 15-video corpus state.

```


17. assistant (2026-07-03T16:41:03.987Z)

Let me check Section 3 to see the analysis commands mentioned:

18. user (2026-07-03T16:41:04.804Z)

```
0      # Alpha Launch Readiness Plan
1
2      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-03
3      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
4      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
5      > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.
6      > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
7
8      ---
9
10     ## 0. TL;DR
11
12     The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before
manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion
schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed
before it can gate anything.
13
14     The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
15
16     ## 1. Problem & goal
17
18     The previous launch blockers were spread across roadmap docs and mixed together three
different questions:
19
20     - **Can an invited tester use the product without SSH?**
21     - **Can the 15-video golden corpus now support honest quality gates?**
22     - **Can we claim or ship an owned/commercial rally detector?**
23
24     This doc separates those into executable workstreams. The goal is to reach an invite-
only alpha where the product is usable end-to-end, the quality claims are honest and
reproducible, and the remaining model/legal gates are explicit rather than accidentally implied
by launch activity.
25
26     ## 2. Decisions locked
27
28     | # | Decision | Source / date | Implication |
29     |---|---|---|---|
30     | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
31     | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
32     | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
33     | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
```

```

windowing safety; keep it as an opt-in/analysis lever. |
34 | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
35
36 ## 3. Verified snapshot, 2026-07-03
37
38 All five Git checkouts were fast-forwarded before analysis:
39
40 | Repo | Branch | Synced HEAD |
41 |---|---|---|
42 | `badminton-highlight-indexer` | `master` | `e4914d1` |
43 | `khelsutra` | `master` | `b0f70a5` |
44 | `rally-annotator` | `main` | `ef9e088` |
45 | `sports-data-collector` | `master` | `bfc014f` |
46 | `sports-obsreport` | `main` | `a0c7252` |
47
48 Corpus/eval facts:
49
50 - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
51 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
52 - `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
53 - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only
6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
54 - `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
55 - `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
56 - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
57 - `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity
group` currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
58
59 ## 4. Launch strategy
60
61 ### 4.1 Alpha lane: make the product usable
62
63 The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
64
65 ### 4.2 Quality lane: make the 15-video corpus gate-ready
66
67 The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
68
69 ### 4.3 Flywheel lane: make tester feedback useful

```

19. assistant (2026-07-03T16:41:06.924Z)

Now let me check the tracker section more carefully to look for dependency issues:

20. user (2026-07-03T16:41:07.492Z)

93 Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
 shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
 row explicitly names a cross-repo batch.

94

95	ID	Deliverable	Repo / owner	Depends on	Gated?	Status	PR
96	----	-----	-----	-----	-----	-----	----
97	A0	Create this alpha readiness tracker and index it in docs	`badminton-highlight-indexer`	-	No	IN PROGRESS	#451
98	A1	Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified	`khelesutra` + `badminton-highlight-indexer`	A0	Owner spend / CF config	TODO	-
99	A2	Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing	`badminton-highlight-indexer`	A0	No	TODO	-
100	A3	Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus	`sports-data-collector` + vault	A2	Owner upload/storage scope	TODO	-
101	A4	Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed	`badminton-highlight-indexer`	A2	Reviewer sign-off	TODO	-
102	A5	Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for future Gen-0 comparisons	`badminton-highlight-indexer`	A2	No	TODO	-
103	A6	Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus	`badminton-highlight-indexer`	A2, A4	No	TODO	-
104	A7	Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15	`badminton-highlight-indexer`	A2	GPU/data availability	TODO	-
105	A8	Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips	`badminton-highlight-indexer`	A6, A7	Reviewer sign-off	TODO	-
106	A9	Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict	`badminton-highlight-indexer`	A5, A8	Owner approval for M0.4 scope	TODO	-
107	A10	Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed	Owner + `badminton-highlight-indexer`	A4, A5, A8	Owner decision	GATED	-
108	A11	Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail	Owner/legal	A10	Legal/provenance	GATED	-
109	A12	Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train reeval` job, and delta-F1 surface	`khelesutra` + `badminton-highlight-indexer`	A1, A3	Owner D13 reopen	GATED	-
110	A13	Alpha launch review: runbook, privacy/consent copy, known limits, rollback/disable steps, and owner GO/NO-GO	Cross-repo	A1, A4, A10, A11	Owner launch decision	TODO	-

111

112 ## 8. Concrete next PR sequence

113

21. assistant (2026-07-03T16:41:09.562Z)

Now let me analyze the tracker dependencies more systematically to check for inefficiencies:

22. user (2026-07-03T16:41:20.979Z)

=== DEPENDENCY ANALYSIS ===

A4: Refresh n=15 nightly regression baseline

- Depends on: A2 only
- For: recording nightly command/output, keeping stale-baseline warnings

A6: Fix backend.eval.ablation import + run R2/tolF1 ablation

- Currently lists: A2, A4
- Actually needs: A2 only (ablation fix doesn't require nightly baseline refresh)
- A4 is gated on 'Reviewer sign-off'

- A6 is gated on 'No' but depends on A4 for no reason

FINDING 1: A6 lists A4 as a dependency but A6 (ablation fix) doesn't actually need A4 (nightly baseline refresh). These should be independent. Currently forces serialization: A2 -> A4 -> A6
Could be: A2 -> {A4, A6} in parallel

=== A8 ANALYSIS ===

A8: Re-run fusion_compare, group ablation, served contrast after A7

- Depends on: A6, A7
- A6 = Fix ablation import, run R2/tolF1 ablation
- A8 wants to run 'group ablation' (unclear if same as R2 ablation)

A6 says 'Fix backend.eval.ablation... then run R2/tolF1 ablation'
A8 says 'run... group ablation...'

POTENTIAL: Both A6 and A8 run ablation() but on different granularities?
If 'group ablation' is --granularity group and R2/tolF1 ablation is different, both could be independent post-A7, but A8 depends on A6 completing first.

=== TRANSITIVE BLOCKING ===

Path: A2 -> A4 -> A6 -> A8 -> A9
A5 (heuristic floor) -> A9
Both A4 and A5 depend on A2
But A4 gates into A6 gates into A8
So A9 waits on this chain even though A5 is independent.

The issue: A6 depends on A4 for no technical reason, creating a serialization chain when A4 and A6 could run in parallel.

23. assistant (2026-07-03T16:41:23.269Z)

Good finding. Let me now check the Section 8 PR sequence for potential re-runs of expensive steps:

24. user (2026-07-03T16:41:24.041Z)

```
112    ## 8. Concrete next PR sequence
113
114    1. **Docs PR:** land this tracker.
115    2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
116    3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.
117    4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
118    5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
119    6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
```

25. assistant (2026-07-03T16:41:26.140Z)

Let me also check if there's duplication in the verification commands in Section 11:

26. user (2026-07-03T16:41:33.166Z)

```
<stdin>:6: SyntaxWarning: invalid escape sequence '\h'
<stdin>:29: SyntaxWarning: invalid escape sequence '\h'
=== SECTION 11 VERIFICATION COMMANDS ===
```

Note: Section 11 uses backslash paths (Windows) while Section 3 uses forward slashes

but this is OK since both sections say '[analysis]' -- the actual runs by the owner

1. fusion_compare: identical in both sections ?
2. serve_contrast: identical --manifest arg ?
3. harness: Section 3 missing --version and --out! These are REQUIRED flags:
Section 3: python -m training.gen0.harness --manifest <rebased-n15> --floor eval_baselines/heuristic_lovo_n6.json
Section 11: python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor eval_baselines/heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
DIFFERENCE: --version and --out missing in Section 3 command!
These are REQUIRED (no defaults) per help text.
4. nightly_regression: identical args ?
5. ablation: identical args ?

=== ISSUE FOUND ===

The Section 3 analysis command is incomplete: missing required --version and --out flags
This would cause the command to fail when run, conflicting with the [analysis] claim that it was successfully run during the 2026-07-03 snapshot.

27. assistant (2026-07-03T16:41:35.158Z)

Perfect. Let me verify this is an actual issue by checking what the help text says about whether --version and --out are required:

28. user (2026-07-03T16:41:38.744Z)

```
usage: harness.py [-h] --manifest MANIFEST [--floor FLOOR] [--out OUT]
                  --version VERSION
```

```
Gen-0 TAL harness (M0.4): ONE command -> train -> honest LOVO eval -> ship-
--
output/human_lovo_manifest.json --floor eval_baselines/heuristic_lovo_n6.json
--out artifacts/rally_tal_gen0 --version 0.1.0 Productionizes the rally_seq
prototype per ADR-008: - TRAINS on the corpus manifest (per-video trajectory +
golden labels), evaluates with honest leave-one-video-out (split by match,
threshold tuned on train only), then fits the FINAL model on all videos. -
Emits a versioned ARTIFACT BUNDLE ? the only thing serving may ever consume:
<out>/<version>/weights.json (gitignored; classifier + frozen normalization)
<out>/<version>/manifest.json (the reviewable contract: feature-schema
version, calibration, training lineage + GT hashes, eval results,
attestations, gate verdict) - Runs the SHIP-GATE as pre-flight: floor
comparison + a PAIRED per-video sign test (at n=6 a mean gap alone is noise-
--
--out OUT
--version VERSION    semver for the artifact bundle
```

29. assistant (2026-07-03T16:41:41.439Z)

Perfect confirmation - both `--version` and `--out` are required (they appear in the usage line without brackets). Now let me check for any other issues with flag names in the verification commands. Let me double-check the ablation command exists with that granularity flag:

30. user (2026-07-03T16:41:44.914Z)

(Bash completed with no output)

31. assistant (2026-07-03T16:41:47.219Z)

The import issue is circular. Let me look at what arguments ablation.py defines when not imported through the broken module:

32. user (2026-07-03T16:41:47.514Z)

```

500     widths = [18, 17, 18, 14, 5, 22]
501
502     def _fmt_row(cells: Sequence[str]) -> str:
503         return " ".join(c.ljust(w) for c, w in zip(cells, widths))
504
505     lines = [header, "", _fmt_row(cols), _fmt_row(["-" * w for w in widths])]
506     for r in rows:
507         ci = f"[{r.ci_low:+.3f}, {r.ci_high:+.3f}]"
508         sign = f"{r.sign_wins}W/{r.sign_losses}L p={r.sign_p_value:.3f}"
509         verdict = VERDICT_LABEL.get(r.verdict, r.verdict)
510         lines.append(
511             _fmt_row(
512                 [
513                     r.signal,
514                     f"{r.delta_f1:+.3f}",
515                     ci,
516                     sign,
517                     "yes" if r.ci_excludes_zero else "no",
518                     verdict,
519                 ]
520             )
521         )
522     return "\n".join(lines)
523
524
525     # ----- PDF
526 export
527     # The ~116-line reportlab PDF builder lives in a sibling module to keep this file a pure
528 driver;
529     # re-exported here so ``ablation.export_pdf`` stays a valid public name. Placed AFTER
530 the
531     # dataclasses/constants it consumes (``ablation_pdf`` imports FROM this module) to avoid
532 a cycle.
533     from backend.eval.ablation_pdf import export_pdf as export_pdf # noqa: E402
534
535     # -----
536 CLI

```

33. user (2026-07-03T16:41:50.658Z)

```

    )
    )
    return "\n".join(lines)

```

----- PDF export

The ~116-line reportlab PDF builder lives in a sibling module to keep this file a pure driver; re-exported here so ``ablation.export\_pdf`` stays a valid public name. Placed AFTER the dataclasses/constants it consumes (``ablation\_pdf`` imports FROM this module) to avoid a cycle.

```
from backend.eval.ablation_pdf import export_pdf as export_pdf # noqa: E402
```

----- CLI

```

def main() -> None:
    ap = argparse.ArgumentParser(
        description="LOS0/LOG0 ablation runner over the golden feature substrate (pure driver "

```

```

    "over experiment.compare + eval_stats; no new scoring math)."
```

)
ap.add_argument(
 "--features-dir",
 default="output",
 help="dir of *_golden_features.json (the persisted golden substrate)",
)
ap.add_argument(
 "--granularity",
 choices=("signal", "group"),
 default="group",
 help="leave-one-SIGNAL-out (column) or leave-one-GROUP-out (producer)",
)
ap.add_argument("--iou", type=float, default=0.5)
ap.add_argument("--seed", type=int, default=0, help="bootstrap seed (determinism)")
ap.add_argument(
 "--litmus",
 action="store_true",
 help="reproduce the static Gate-A measurement (CONTROL vs min_crossings=2) "
 "instead of the learned full-set ablation",
)
ap.add_argument(
 "--pdf", default=None, help="also write a PDF ablation report to this path"
)
ap.add_argument(
 "--json", default=None, help="also write the structured report JSON here"
)
args = ap.parse_args()

videos = load_videos(args.features_dir)
if len(videos) < 2:
 raise SystemExit(
 f"need >=2 *_golden_features.json in {args.features_dir} "
 f"(found {len(videos)})"
)

if args.litmus:
 cfg = litmus_config()
 cfg.iou, cfg.seed = args.iou, args.seed
else:
 cfg = AblationConfig(
 full=learned_full_variant(),

34. assistant (2026-07-03T16:41:53.067Z)

Good, the `--granularity` flag exists with choices "signal" and "group", and the default is "group". Let me now summarize all findings:

35. user (2026-07-03T16:42:03.468Z)

```

[
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 103,
    "summary": "A6 lists A4 as a dependency, but ablation import fix does not actually require nightly baseline refresh. Forces unnecessary serialization: A2->A4->A6 when A4 and A6 could run in parallel post-A2.",
    "failure_scenario": "A4 blocks A6 (via A2->A4->A6 chain) even though baseline refresh and ablation fix are independent work. This delays the ablation fix by the time needed for baseline refresh + reviewer sign-off, which could take days."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 55,
    "summary": "Section 3 analysis narrative claims successful gen0.harness run but the command

```

shown is incomplete: missing required --version and --out flags. Section 11 shows the complete command with these flags.",

"failure_scenario": "If someone tries to reproduce the 2026-07-03 snapshot using the Section 3 command, it will fail with 'required argument missing' error. The [analysis] claim contradicts the executable command shown on the same page."

```
},
{
  "file": "docs/ALPHA_LAUNCH_READINESS.md",
  "line": 57,
  "summary": "Ablation import circular dependency (backend/eval/ablation.py line 530 imports from backend/eval/ablation_pdf.py, which imports from ablation.py) causes 'python -m backend.eval.ablation' to fail immediately. A6 depends on fixing this, but A6 also depends on A4 which is gated on reviewer sign-off, delaying the fix.",
  "failure_scenario": "Running the verification command from line 164 now fails with ImportError. The circular import blocks all ablation work until A6 is executed, but A6 is serialized behind A4 (nightly baseline refresh + reviewer review).",
},
{
  "file": "docs/ALPHA_LAUNCH_READINESS.md",
  "line": 105,
  "summary": "A8 depends on A6 (fix ablation + run R2/tolF1 ablation) but A8 also plans to 'run group ablation'. If these are different ablation commands (group vs tolF1 granularity), A8 waits for A6's tolF1 ablation completion before running group ablation, even though they could run in parallel post-A7.",
  "failure_scenario": "If A6 runs a time-consuming R2/tolF1 ablation and A8 later runs a separate group ablation with the same data, the second run re-executes work that could have been batched. This is expensive (many hours GPU time per the doc context) and could be run in parallel if both depend only on A7 and the ablation import fix.",
},
{
  "file": "docs/ALPHA_LAUNCH_READINESS.md",
  "line": 162,
  "summary": "Section 11 reference command for gen0.harness is complete but does not match the Section 3 narrative claim. The placeholder '--version analysis-n15' and '--out <temp-out>' show a different invocation than what the snapshot text implies was a single definitive run.",
  "failure_scenario": "If the doc is used as a reproduction guide, the mismatch between narrative (Section 3: 'produced learned LOVO mean F1 0.551') and command (Section 11: only shows one version) raises doubt about which exact command produced those results, complicating future audit trails."
}
]
```

36. assistant (2026-07-03T16:42:16.071Z)

```
```json
[
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 103,
 "summary": "A6 lists A4 as a dependency, but ablation import fix does not actually require nightly baseline refresh. Forces unnecessary serialization: A2->A4->A6 when A4 and A6 could run in parallel post-A2.",
 "failure_scenario": "A4 blocks A6 (via A2->A4->A6 chain) even though baseline refresh and ablation fix are independent work. This delays the ablation fix by the time needed for baseline refresh + reviewer sign-off, which could take days."
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 55,
 "summary": "Section 3 analysis narrative claims successful gen0.harness run but the command shown is incomplete: missing required --version and --out flags. Section 11 shows the complete command with these flags.",
 "failure_scenario": "If someone tries to reproduce the 2026-07-03 snapshot using the Section 3 command, it will fail with 'required argument missing' error. The [analysis] claim contradicts the executable command shown on the same page."
 }
]
```



```

},
{
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 57,
 "summary": "Ablation import circular dependency causes 'python -m backend.eval.ablation' to fail immediately. A6 depends on fixing this, but is serialized behind A4 (nightly baseline refresh + reviewer review), delaying the fix.",
 "failure_scenario": "Running the verification command from line 164 now fails with ImportError. The circular import blocks all ablation work until A6 is executed, but A6 is serialized behind A4 review gate."
},
{
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 105,
 "summary": "A8 depends on A6 (fix ablation + run R2/tolF1 ablation) but plans to 'run group ablation'. If these are different granularities, expensive ablation work may be re-run when both could execute in parallel post-A7.",
 "failure_scenario": "A6 runs ablation with one granularity, then A8 re-runs ablation with a different granularity, re-executing expensive GPU analysis that could have been batched together if both depended only on A7 and the ablation import fix."
}
]
...

```